

Aquaculture Intelligence System (AIS)

Complete Dataset Specifications & Data Types Catalogue

Document Title:	Dataset Specifications & Data Types Catalogue — AIS
Data Modalities:	Tabular Sensor Streams, Image (RGB/Labels), Text (PDF), GeoTIFF Rasters, Vector Embeddings
Total Datasets Catalogued:	22 Primary & Auxiliary Datasets across 6 AI Modules
Storage Engines:	TimescaleDB Hypertables, PostgreSQL 16, pgvector HNSW, MinIO S3 Blob Storage
Data Licensing:	Public Domain, CC BY 4.0, CC BY-NC, Govt Open Data, Self-Generated Synthetic

SUMMARY OF SYSTEM DATA TYPES & MODALITIES:

- **Tabular Time-Series (Sensors & Feeds):** High-frequency IoT streams (DO, pH, Temp, NH3, Turbidity) stored in TimescaleDB 7-day hypertable chunks.
- **Computer Vision Images (Disease & Batch):** 224x224 RGB photos & 640x640 bounding box annotations stored in MinIO S3 object storage.
- **Unstructured PDF Text (RAG Corpus):** Aquaculture manuals chunked into 400-token blocks with source section metadata.
- **Dense Vector Embeddings (RAG Vectors):** 384-dimensional float arrays stored in pgvector with HNSW cosine-similarity index.
- **Multispectral Satellite Rasters (Remote Sensing):** 6-band Sentinel-2 GeoTIFF tiles (B02, B03, B04, B08, B8A, B11) for NDWI & Chlorophyll-a.

1. Complete Master Dataset Catalogue

Module	Dataset Name	Source / Provider	Data Type & Format	Volume	License
Water Quality	UCI Water Quality	UCI ML Repository	Tabular (CSV)	3,276 rows	Public Domain
Water Quality	USGS NWIS Sensor Data	waterdata.usgs.gov	Tabular (JSON/CSV)	5M+ rows	Public Domain
Water Quality	NOAA Aqua Monitoring	noaa.gov	Tabular (CSV)	~50K records	Public Domain
Water Quality	Synthetic Sensor Streams	ARIMA + Noise Gen	Tabular (CSV)	500K records	Self-Generated
Fish Disease	Kaggle Fish Disease	Kaggle Community	Image (JPEG)	~9,000 imgs	CC BY 4.0
Fish Disease	Roboflow Aquaculture	roboflow.com	Image + YOLO Labels	~3,500 imgs	CC BY 4.0
Fish Disease	FishNet Open Images	fishnet.ai	Image (JPEG)	~94,000 imgs	CC BY-NC
Fish Disease	Na Rang Nong Khai	Thai Dept Fisheries	Image (JPEG)	~2,100 imgs	Research Use
Fish Disease	Gen AI Synthetic Aug	Claude API Image Gen	Image (JPEG)	~1,500 imgs	Self-Generated
Fish Disease	YOLOv8 Batch Net-Pull	Manual CVAT Annot	Image + YOLO Label	~500 imgs	Self-Generated
Growth / Feed	FAO Fishstat	fao.org/fishery	Time-Series (CSV/XLSX)	50+ yr series	CC BY-NC
Growth / Feed	ICAR-CIFA SGR Tables	cifa.nic.in	Tabular / Lookup Matrix	Published PDF	Govt Open Data
Growth / Feed	NACA Pond Trial Data	enaca.network	Tabular (PDF/CSV)	~10K trials	CC BY-NC
Growth / Feed	WorldFish GIFT Data	worldfishcenter.org	Tabular (CSV)	Tilapia trials	Research Use
Growth / Feed	Synthetic Growth Curves	Von Bertalanffy Sim	Tabular (CSV)	500K curves	Self-Generated

Gen AI RAG	ICAR-CIFA Tech Bulletins	cifa.nic.in	Document (PDF)	25+ PDFs (~600p)	Govt Open Data
Gen AI RAG	NACA Health Guides	enaca.network	Document (PDF)	12 manuals (~400p)	CC BY-NC
Gen AI RAG	FAO Fisheries Papers	fao.org	Document (PDF)	8 vols (~500p)	CC BY-NC
Gen AI RAG	MPEDA Export Docs	mpeda.gov.in	Document (PDF)	5 docs (~80p)	Govt Open Data
Gen AI RAG	State Scheme Docs	State Portals	Document (PDF)	10 docs (~120p)	Public Domain
Satellite	Sentinel-2 L2A Imagery	Copernicus Hub	GeoTIFF (6-Band 10m)	Unlimited API	CC BY (Copernicus)
Satellite	AquaVision Shrimp Ponds	Research Consortium	GeoTIFF + Shapefiles	~2K tiles	Research Use
Satellite	SpaceNet Water Bodies	spacenet.ai	GeoTIFF + GeoJSON	~10K tiles	CC BY-SA 4.0
Satellite	MODIS Aqua Chlorophyll	NASA OceanColor	HDF / NetCDF (1km)	Daily global	Public Domain

2. Data Modality Preprocessing & Schema Mapping

A. Tabular Sensor Stream Modality

Storage Engine: TimescaleDB Hypertable (sensor_readings)
Features Extracted: 15m/1h/6h rolling means, rolling standard deviation, diurnal cyclical sine/cosine encodings (hour of day, day of year), and t-1m, t-1h, t-24h lag variables.
Validation & Cleaning: Hard bound range validation (DO [0,20], pH [3,11], Temp [5,45], NH3 [0,5]), Z-score spike filtering ($|Z| > 3.5$ replaced by local median), forward-fill (<10 min gap) and linear interpolation (10-60 min gap).

B. Computer Vision Image Modality

Storage Engine: MinIO S3 Object Storage (images/raw/ & images/processed/)
Preprocessing Pipeline: Resolution check ($\geq 64\text{px}$), OpenCV Laplacian variance blur filter (blur_score < 80 rejected), RGB color space standardization, ImageNet mean/std normalization (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).
Augmentation Strategy: Albumentations (RandomHorizontalFlip, RandomRotation(20), ColorJitter, Cutout) and Claude API synthetic image generation for rare disease classes (<200 samples).

C. Unstructured PDF Text & Vector Embedding Modality

Storage Engine: PostgreSQL 16 with pgvector extension (rag_chunks hypertable)
Text Extraction & Chunking: PyMuPDF column layout text extraction, header/footer repetition stripping, RecursiveCharacterTextSplitter (chunk_size=400 tokens, overlap=50 tokens).
Embedding Model & Indexing: sentence-transformers/all-MiniLM-L6-v2 (384-dimensional dense vectors), indexed via HNSW graph algorithm (m=16, ef_construction=64) for ultra-fast sub-50ms cosine similarity query retrieval.

D. Multispectral Satellite GeoTIFF Modality

Storage Engine: Sentinel Hub API + MinIO S3 Tile Cache
Spectral Band Processing: 6 Sentinel-2 L2A bands extracted at 10m spatial resolution (B02 Blue, B03 Green, B04 Red, B08 NIR, B8A RedEdge, B11 SWIR1). Reflectance scaling factor division by 10,000.
Index Calculation: Normalized Difference Water Index NDWI = $(B03 - B08) / (B03 + B08)$, Chlorophyll-a proxy = $B03 / B04$, Turbidity proxy = $B04 / B03$. ResNet34 U-Net 4-class land cover segmentation (water, vegetation, soil, structure).